Name: Oscar Cortez
ITAI 4375 – AI in Healthcare
Professor Doreen Rosenstrauch
February 18, 2026

Prehospital Early Warning Scores to Predict 30-Day Mortality

This report reviews a study by Lindskou and colleagues (2023) that looked at several prehospital Early Warning Scores (EWSs). The goal was to see how well these scores can predict whether an adult ambulance patient will die within 30 days using the first vital signs taken on scene. The study also looked at 1-day mortality and ICU admission (Lindskou et al., 2023).

This study did not create a new model. Instead, it tested and compared scoring tools that already exist. The scores are based on basic EMS information such as respiratory rate, oxygen saturation, blood pressure, heart rate, temperature (in some scores), and level of consciousness (Lindskou et al., 2023). Since EMS already collects these vital signs, using these scores in the field is realistic.

The results showed moderate accuracy for predicting 30-day mortality. The best score (mNEWS) had an AUROC of 0.68 (95% CI 0.68–0.69). The qSOFA score did worse (AUROC 0.59) (Lindskou et al., 2023). Precision-recall results were also low (AUPRC about 0.09–0.14), which matters because death was not common in this group. This means the scores can flag some high-risk patients, but they also create many false alarms. In real EMS systems, that can lead to overtriage and extra use of resources.

One strength of the study is the large real-world sample (107,569 patients) and the fact that outcomes were linked to mortality data (Lindskou et al., 2023). Another strength is that the scores use objective data that EMS already collects. However, the scores are limited because their accuracy is only moderate and they can produce many false positives at common cutoffs. This can cause unnecessary alerts or “alarm fatigue.” Another limitation is that the study comes

from one regional EMS system, so the results may not match other systems with different protocols or patient populations. Also, these scores are simple point systems, so they may miss more complex patterns that newer machine-learning models can sometimes detect. Overall, EMS crews and receiving hospitals may benefit because the scores help standardize communication and support early risk recognition. Still, they should be used to support decisions, not replace clinical judgment. The scores provide a helpful warning signal, but they are not strong enough to be used alone (Lindskou et al., 2023).

References

Guan, G., Lee, C. M. Y., Begg, S., Crombie, A., & Mnatzaganian, G. (2022). The use of early warning system scores in prehospital and emergency department settings to predict clinical deterioration: A systematic review and meta-analysis. *PLOS ONE, 17*(3), e0265559. https://doi.org/10.1371/journal.pone.0265559

Lindskou, T. A., Ward, L. M., Søvsø, M. B., Mogensen, M. L., & Christensen, E. F. (2023). Prehospital early warning scores to predict mortality in patients using ambulances. *JAMA Network Open, 6*(8), e2328128. https://doi.org/10.1001/jamanetworkopen.2023.28128